

Technical Analysis of IPv4 Header Fields and Network Layer Protocols

Summary

This document explains critical IPv4 header fields—**Time to Live (TTL)**, **Protocol**, and **Header Checksum**—and explains how they enable routing control, loop prevention, congestion handling, and protocol demultiplexing.

Key insights:

- **TTL prevents infinite routing loops** through hop-count limitation.
- **Protocol field identifies upper-layer protocol** (e.g., TCP, UDP).
- **Header Checksum is recalculated at every hop** due to header modifications.
- Certain header fields remain constant, while others change during transit.
- Hexadecimal parsing requires understanding bit-to-hex mapping.

1. Time to Live (TTL) Field

1.1 Field Characteristics

- Size: 8 bits
- Range: 0–255 ($2^8 - 1$)
- Initialized by source host (never set to 0 initially)

1.2 Decrement Mechanism

- Each router decrements TTL by exactly 1.
- TTL represents hop count, not time in seconds.

1.3 Discard Conditions

At a Router

- If TTL becomes 0 at a router:
 - Packet is discarded.
 - Router sends ICMP "TTL Exceeded" message to source.

At the Destination

- If TTL reaches 0 exactly upon arrival at destination:
 - Packet is accepted.
 - No further forwarding required.

1.4 Functional Purpose

TTL:

- Prevents infinite looping.
- Conserves bandwidth.
- Enables tools like traceroute (by observing TTL expiry behavior).

2. Protocol Field

2.1 Functional Role

- Size: 8 bits
- Identifies encapsulated upper-layer protocol.
- Allows correct delivery to Transport or Network-layer protocol handler.

Common protocol numbers:

Protocol	Decimal Value
ICMP	1
IGMP	2
TCP	6
UDP	17
OSPF	89

2.2 Buffer Management and Discard Priority

When router buffers are full, selective discarding may occur.

Typical priority order (lowest to highest):

1. ICMP (1)

2. IGMP (2)
3. UDP (17)
4. TCP (6)

Reason:

- TCP is connection-oriented.
- Dropping TCP packets may trigger retransmissions and congestion control reactions.
- Control protocols (ICMP/IGMP) are less critical to session continuity.

3. Header Checksum

3.1 Purpose

- Size: 16 bits
- Detects errors in IPv4 header only.
- Does not protect payload data.

Payload protection is handled by:

- TCP checksum
- UDP checksum

3.2 Recalculation Requirement

Header checksum must be recalculated at every router because:

- TTL changes at each hop.
- Fragmentation may modify:
 - Total Length
 - Flags
 - Fragment Offset
- Options field may change.

Any header modification invalidates previous checksum.

4. Header Field Stability During Transit

4.1 Fields That Never Change

- Version
- Services (ToS)
- Identification
- DF flag
- Protocol
- Source IP
- Destination IP

4.2 Fields That May Change (Conditional)

- Header Length (if options used)
- Total Length (during fragmentation)
- MF flag
- Fragment Offset
- Options

4.3 Fields That Definitely Change

- TTL (decremented every hop)
- Header Checksum (recomputed every hop)

5. IPv4 Options Field

5.1 Size Characteristics

- Range: 0–40 bytes
- Maximum total header size: 60 bytes

5.2 Common Options

Record Route

- Records router IP addresses.
- Max space: 40 bytes

- 2 bytes overhead (Type + Length)
- Remaining 38 bytes
- Each IP = 4 bytes
- Maximum ≈ 9 router addresses

Strict Source Routing

- Packet must follow exact specified path.

Loose Source Routing

- Packet must visit specified routers.
- Intermediate routers allowed.

Timestamp

- Routers record arrival time.
- Used to compute delays.

Padding

- Ensures header ends on 32-bit boundary.
- Required if options do not align properly.

6. IPv4 Header Mnemonic Structure

Mnemonic:

Very Handsome Student Is Fully Fine To Play Hockey

Row 1

- Version
- Header Length
- Services
- Total Length

Row 2

- Identification
- Flags
- Fragment Offset

Row 3

- Time to Live
- Protocol
- Header Checksum

7. Hexadecimal Header Parsing Methodology

7.1 Bit-to-Hex Mapping

- 1 hex digit = 4 bits
- 8 hex digits = 32 bits (1 header row)

7.2 Row Structure in Hex

- First 8 hex digits → Row 1
- Next 8 hex digits → Row 2
- Next 8 hex digits → Row 3

7.3 Extracting TTL and Protocol

In Row 3:

- First 2 hex digits → TTL (8 bits)
- Next 2 hex digits → Protocol (8 bits)

Example

If Row 3 begins:

5906

TTL:

59 (hex)
= 89 (decimal)

Packet can traverse 89 more hops.

Protocol:

06 (hex)
= 6 (decimal)

Encapsulated protocol:

TCP

